	GLOBAL ECONOMICS **GRADE: 10TH** **MIDTERM TASK** **TEACHER'S NAME: Nicolás López Cuéllar**	**3 TERM** **2025–2026**

Learning objective: Use continuous probability distributions and normal distributions to model contexts, calculate probabilities, draw distributions, and find cutoff values.	**Assessment criteria** C1. Online Coffee Sales Investment. C2. Three-Scenario Inventory Investment. C3. Small-Business Investment. C4. Drawing and comparing probability distributions. C5. Uses the standard normal distribution and its tables to calculate probabilities. C6. Uses normal distributions in contextual situations. C7. Uses inverse normal reasoning to find values and intervals.

Criteria assessment

This task sheet assesses criteria C1 to C7. Each problem focuses on the criterion shown in its title. The number of points available for each task appears at the beginning of the item. Partial credit may be awarded when the procedure is correct but there are arithmetic errors. Answers without procedure may not receive full credit.

1. Online Coffee Sales Investment (C1)

A small entrepreneur runs a specialty coffee resale business. Let V be the end-of-quarter value, measured in million pesos. The final value cannot be below 20 and cannot exceed 68. Lower values are possible, but higher values become gradually more likely across this interval, so the final value is modeled with an increasing density. The initial investment is 32 million pesos. The probability of the extreme value of 20 million pesos can be considered to be 0.

a) **(3)** Determine the density constant and write the pdf.

b) **(2)** Find the probability of loss and the probability of earning.

c) **(4)** Compute the probability that the money obtained falls between 32 and 56 million pesos.

2. Three-Scenario Inventory Investment (C2)

A seller invests in a short-term inventory deal to resell packaged foods in nearby cities. For the quarter, they consider 3 realistic market scenarios. In Scenario 1, transport bottlenecks continue and restocking stays slow; this has probability 0.2. In Scenario 2, transport returns to normal and inventory rotates faster; this has probability 0.5. In Scenario 3, demand increases because restaurants and small stores begin buying in bulk; this has probability 0.3. If Scenario 1 occurs, the final value of the investment is expected to end between 48 and 68 million pesos, with all values in that interval treated as equally plausible. If Scenario 2 occurs, the final value is expected to end between 54 and 104 million pesos, again with all values in that interval treated as equally plausible. If Scenario 3 occurs, the final value is expected to end between 60 and 90 million pesos, again with all values in that interval treated as equally plausible. The initial investment is 60 million pesos.

a) **(2)** Write the conditional density for each scenario.

b) **(3)** Build the unconditional pdf of the final value as a piecewise function.

c) **(4)** Find the probability of loss and the probability of earning.

3. Small-Business Investment (C3)

An investor invests in a short-term small-business opportunity: buying low-cost clothing bundles to resell online and at weekend markets. They estimate the final value cannot be below 0 million pesos because, in this simplified model, the end-of-period value cannot be negative. They also estimate it cannot exceed 40 million pesos because their first-round sales capacity is limited. Their most likely final value is 16 million pesos, based on expected demand from regular customers. Values below 16 million pesos are possible, but become less likely as results move farther down from that level. Values above 16 million pesos are also possible, but after that point, high outcomes become gradually less likely. Since the model is continuous, the probability of obtaining any exact extreme value can be considered 0, so the probabilities of ending at exactly 0 million pesos or exactly 40 million pesos are treated as 0. The initial investment is 10 million pesos.

- **a)** **(3)** Write the pdf as a piecewise function.
- **b)** **(3)** Find the probability of loss and the probability of earning.
- **c)** **(3)** Find the specific probability of earning more than 50% of the investment.

4. Probability Distribution Graphs (C4)

- **a)** **(3)** Draw four uniform distributions on the same axes.

$$A : (\mu = -1, \sigma \approx 0.58), \quad B : (\mu = 2, \sigma \approx 1.15)$$
$$C : (\mu = 6, \sigma \approx 1.73), \quad D : (\mu = 9, \sigma \approx 2.31)$$

- **b)** **(3)** Draw four centered triangular distributions on the same axes.

$$A : (\mu = -1, \sigma \approx 0.82), \quad B : (\mu = 2, \sigma \approx 1.22)$$
$$C : (\mu = 6, \sigma \approx 0.41), \quad D : (\mu = 9, \sigma \approx 1.63)$$

- **c)** **(3)** Draw four normal distributions on the same axes.

$$A : X \sim N(1.0, 1.2^2), \quad B : Y \sim N(3.0, 0.7^2)$$
$$C : Z \sim N(5.0, 1.6^2), \quad D : W \sim N(7.0, 1.0^2)$$

5. Standard Normal Table Problems (C5)

Let $Z \sim N(0, 1)$. Let $\Phi(z) = P(Z < z)$ for $z \in [0, \infty)$.

- **a)** **(4)** Calculate the following standard-normal probabilities:

$$P(Z < 1.50), \quad P(Z < -1.20)$$
$$P(Z < 0.25), \quad P(Z < 1.15)$$
$$P(Z < -1.05), \quad P(Z < 1.35)$$

- **b)** **(2)** Calculate the following standard-normal probabilities:

$$P(Z > 1.50), \quad P(Z > -0.80), \quad P(Z > 1.35)$$

- **c)** **(4)** Calculate the following standard-normal probabilities:

$$P(0.25 < Z < 1.15), \quad P(Z < -1.05 \text{ or } Z > 1.35)$$

$$P(Z < -1.28), \quad P(Z < 0.84)$$
$$P(Z > 1.28), \quad P(Z > -0.84)$$

$$P(-1.96 < Z < 1.96), \quad P(Z < -1.64 \text{ or } Z > 1.64)$$

6. Contextual Normal Distribution Problems (C6)

- **a)** **(2)** If exam scores follow $X \sim N(70, 8^2)$, find the probability that a student scores below 82 and the probability that a student scores below 60.4: $P(X < 82)$ and $P(X < 60.4)$.
- **b)** **(2)** If delivery times follow $X \sim N(56, 5^2)$, find the probability that a delivery takes more than 63.5 minutes and the probability that a delivery takes more than 52 minutes: $P(X > 63.5)$ and $P(X > 52)$.
- **c)** **(3)** If weekly app usage follows $X \sim N(14, 4^2)$, find the probability that weekly app usage is between 15 and 18.6 hours: $P(15 < X < 18.6)$.
- **d)** **(3)** If scholarship scores follow $X \sim N(300, 20^2)$, find the probability that a score is below 279 or above 327: $P(X < 279 \text{ or } X > 327)$.

7. Inverse Normal Problems (C7)

- **a)** **(2)** If commute times follow $X \sim N(38, 7^2)$, find the cutoff below which the lowest 10% of commutes fall and the cutoff below which the lowest 80% of commutes fall: $P(X < x) = 0.10$ and $P(X < x) = 0.80$.
- **b)** **(2)** If exam scores follow $X \sim N(72, 9^2)$, find the cutoff above which the highest 10% of scores fall and the cutoff above which the highest 80% of scores fall: $P(X > x) = 0.10$ and $P(X > x) = 0.80$.
- **c)** **(3)** If battery life follows $X \sim N(20, 2.5^2)$, find the interval $[a, b]$ containing the middle 95% of battery lives: $P(a < X < b) = 0.95$.
- **d)** **(3)** If metal rod diameters follow $X \sim N(10.00, 0.08^2)$, find the two cutoff values a and b so that 10% of values are outside the interval: $P(X < a \text{ or } X > b) = 0.10$.